



Attendexa



كلية علوم وهندسة الحاسب الآلي
College of Computer Science and Engineering
جامعة حفر الباطن University of Hafer Al Batin

Attendexa:

A Secure Anonymous Smart Attendance Platform for Universities

By: Ghaida Bandar Aldossary	2231004728
Reem Mohammed aba alkhayl	2231005763
Almas Racheed Alanazi	2231006033

Supervisor: Dr. Mazyona Al-Fagham
April 23th, 2026





Outline

- ☒ **Foreword**

✓ -----

- ☐ Background
- ☐ Research Problem
- ☐ Contribution
- ☐ Related Work

- ☐ Design and Implementation
- ☐ Results
- ☐ Conclusion
- ☐ Time table
- ☐ Future Work





Foreword

ATTENDEXA is a secure smart attendance platform designed for university environments. The project focuses on solving common attendance problems such as proxy attendance, manual errors, wasted lecture time, and weak data protection.

The system uses multiple verification layers to make sure that the student who records attendance is actually present in the correct classroom. It also provides dashboards for students, instructors, and administrators to make attendance management faster, clearer, and more reliable.





Outline

- ☐ Foreword
- ☒ **Background**
 - ✓ -----
- ☐ Research Problem
- ☐ Contribution
- ☐ Related Work
- ☐ Design and Implementation
- ☐ Results
- ☐ Conclusion
- ☐ Time tabel
- ☐ Future Work





Attendexa



Background

Attendance is critical in universities as it directly affects academic performance and institutional records. However, traditional methods like manual roll calls or static QR codes are slow and easily manipulated **as students can sign for one another or share QR codes via messaging apps. Additionally, hardware-based alternatives, such as card-scanning machines, are highly expensive and impractical.** Consequently, these limitations lead to inaccurate tracking and increased workload for instructors.



Attendance Check-in

Scan the QR code displayed in your classroom

● Session is open — CS301 Data Structures (B-201)



Network
University Wi-Fi ✓

Geofence
Checking...

QR Code
Refreshes every 10s

If you lose connection, your check-in will be saved locally for up to 60 seconds and submitted automatically.





Attendexa



Outline

- ☐ Foreword
- ☐ Background
- ☐ **Research Problem**
- ☐ Contribution
- ☐ Related Work
- ☐ Design and Implementation
- ☐ Results
- ☐ Conclusion
- ☐ Time table
- ☐ Future Work



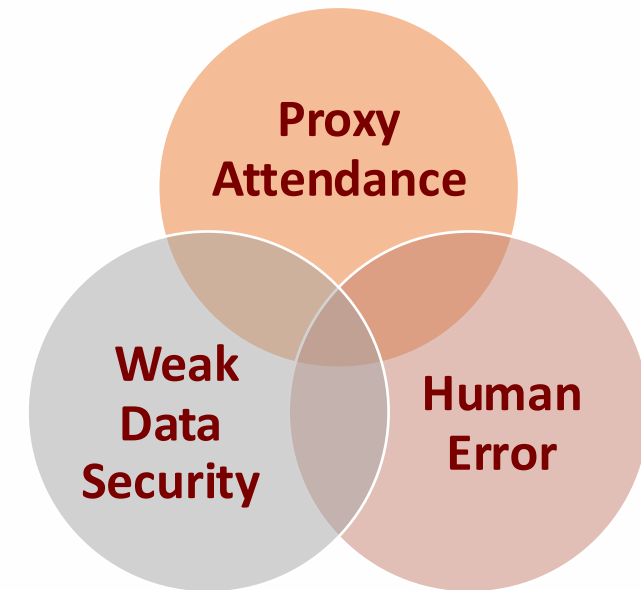


Research Problem

Current attendance systems fail to verify the **physical presence** of students in the classroom.

Key Issues Addressed:

- 1- **Proxy Attendance:** Eliminating the use of shared QR codes or manual signatures for absent classmates.
- 2- **Administrative Burden:** Reducing human errors and the time consumed by manual tracking.
- 3- **Data Vulnerability:** Strengthening security, privacy, and audit logs to protect student information.





Attendexa

كلية علوم وهندسة الحاسب الآلي
College of Computer Science and Engineering
جامعة حفر الباطن
University of Hafer Al Batin



Project Overview & Demo

جامعة حفر الباطن
University of Hafr Al Batin





Outline

- ☐ Foreword
- ☐ Background
- ☐ Research Problem
- ☐ **Contribution**
 - ✓ Goals
 - ✓ Motivations
 - ✓ Findings
- ☐ Related Work
- ☐ Design and Implementation
- ☐ Results
- ☐ Conclusion
- ☐ Time table
- ☐ Future Work





Contribution: Goals



Zero Proxy

Eliminate identity fraud via multi-layer verification.



Automation

Minimize manual effort and save lecture time.



Visibility

Real-time access to records for students and staff.



Security

Ensuring privacy through advanced encryption.



Integration

Seamless connection with University systems.



Analytics

Precise reporting and AI-driven insights.



Contribution: Motivations



Fairness

Ensure fair attendance tracking for all students.



Accuracy

Deliver accurate and reliable attendance records.



Privacy

Protect student data with privacy-first design.



Digital Transformation

Support modern and efficient university operations.





Attendexa

كلية علوم وهندسة الحاسب الآلي
College of Computer Science and Engineering
جامعة حفر الباطن
University of Hafer Al Batin



Contribution: Findings

Our multi-layer attendance system delivers stronger protection than traditional methods.



Dynamic QR Codes

Reduce screenshot sharing.



GPS Geofencing

Confirms the classroom location.



University Wi-Fi Validation

Adds another location-check layer.



Device-Bound Sessions

Reduce account sharing.



Real-Time Dashboards

Help students track absences early.



Admin Analytics

Identify attendance patterns and possible fraud cases.

Overall Impact



Stronger Security
with multi-layer verification.



Save Time
for instructors and students.



Better Experience
for students, instructors, and administrators.



Accurate Records
and data-driven decisions.





Outline

- ☐ Foreword
- ☐ Background
- ☐ Research Problem
- ☐ Contribution
- ☐ **Related Work**
- ☐ Design and Implementation
- ☐ Results
- ☐ Conclusion
- ☐ Time table
- ☐ Future Work





Related Work

Feature / Capability	SAMSTM (Related Work)	Attendexa (Our System)
Verification Method	Single Scan (Static QR)	Multi-Layer (QR + GPS + Wi-Fi)
QR Code Security	Static (Fixed Image)	Dynamic (Refreshes every 10s)
Location Validation	None (Device Independent)	GPS Geofencing (Classroom level)
Network Layer Check	Standard Internet	University Wi-Fi Validation
Proxy Protection	Low (Vulnerable to sharing)	High (Hardware ID + Location)
Fraud Detection AI	Not Available	Real-time Analytics & Alerts

Reference: Ng Sam Kee (2021). "Student Attendance System Based on QR Code with Mobile Devices".





Outline

- ☐ Foreword
- ☐ Background
- ☐ Research Problem
- ☐ Contribution
- ☐ Related Work
- ☐ **Design and Implementation**
 - ✓ System Design
 - ✓ Implementation
- ☐ Results
- ☐ Conclusion
- ☐ Time table
- ☐ Future Work





System Design

ATTENDEXA is designed as a web-based platform with three main portals:

1.Student Portal

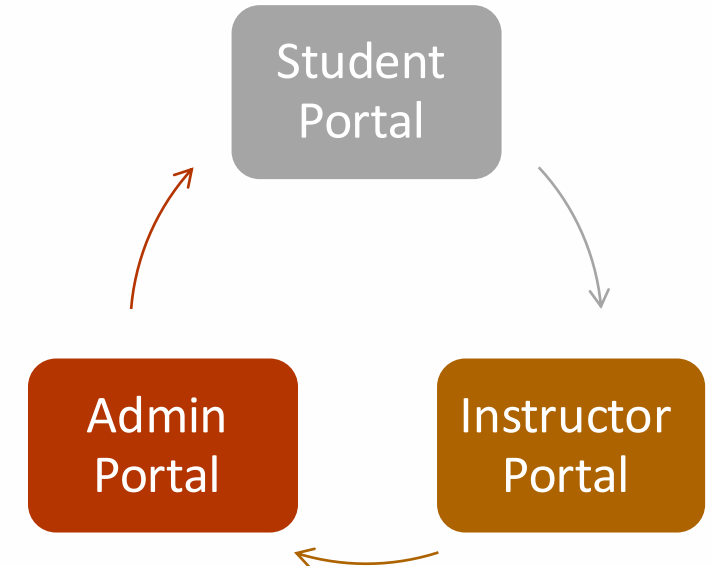
Students can log in, view attendance records, check course absence percentages, receive alerts, and scan QR codes to check in.

2.Instructor Portal

Instructors can create attendance sessions, display dynamic QR codes, monitor live attendance, manage sessions, and export attendance data.

3.Admin Portal

Administrators can manage users, rooms, GPS locations, reports, and AI analytics.





System Design(count..)

Fronted Architecture:



React & Vite

High-performance Single Page Application (SPA) with lightning-fast hot module replacement.



TypeScript

Rigorous type safety ensuring code maintainability and reducing runtime exceptions across portals.



Tailwind & Shadcn

A modular design system providing accessible components and responsive layouts for all devices.





System Design(count..)

Backend Architecture:

-  **Node.js & Express:** Scalable, event-driven RESTful API handling high-concurrency check-in sessions.
-  **PostgreSQL DB:** Robust relational management for users, sessions, and encrypted attendance logs.
-  **JWT & Bcrypt:** Industry-standard stateless authentication and cryptographic credential hashing.





System Design(count..)

10- Seconds TOTP Logic:

To eliminate "Proxy Attendance" (screenshot sharing), the system implements a custom Hash-based Verification Code (HOTP/TOTP).

```
QR_Payload = HMAC-SHA256(Secret,  
Sub_Session_ID, T_Now)
```

This ensures the payload is valid only within a 10-second window, preventing static photo reuse.





Attendexa

كلية علوم وهندسة الحاسب الآلي
College of Computer Science and Engineering
جامعة حفر الباطن
University of Hafer Al Batin



System Implementation



Attendexa

University Attendance Management System

Sign In

Enter your university email to continue

University Email

name@university.edu

Send Verification Code

Demo accounts: admin@ / instructor@ / student@university.edu

Protected by AES-256 encryption · TLS secured · Device-bound sessions



Outline

- ☐ Foreword
- ☐ Background
- ☐ Research Problem
- ☐ Contribution
- ☐ Related Work
- ☐ Design and Implementation
- ☐ **Results**
- ☐ Conclusion
- ☐ Time table
- ☐ Future Work





Results



Student Dashboard

- Real-time Absence Tracking
- Late Arrival Monitoring
- Proactive Status Alerts



Instructor Portal

- Secure, Dynamic QR Code (AES-256)
- Automated Attendance Logs (P/A/L)
- Live Student Monitoring



Admin Oversight

- System-wide AI Insights
- Video & Location Fraud Analytics
- Advanced Security Alerts



Outline

- ☐ Foreword
- ☐ Background
- ☐ Research Problem
- ☐ Contribution
- ☐ Related Work
- ☐ Design and Implementation
- ☐ Results
- ☐ **Conclusion**
- ☐ Time table
- ☐ Future Work



Conclusion

ATTENDEXA provides a secure and modern solution for university attendance management. It replaces manual and weak attendance methods with a multi-layer verification system that improves accuracy, fairness, and security.

The system benefits students by giving them real-time attendance visibility. It benefits instructors by reducing manual work and saving lecture time. It benefits administrators by providing accurate reports, room management, and fraud detection insights.

Overall, ATTENDEXA supports digital transformation in education and creates a more trustworthy attendance process.





Outline

- ☐ Foreword
- ☐ Background
- ☐ Research Problem
- ☐ Contribution
- ☐ Related Work
- ☐ Design and Implementation
- ☐ Results
- ☐ Conclusion
- ☐ **Time table**
- ☐ **Future work**





Timetable

Week	Main Task	Description
Weeks 1–4	Requirements Analysis	Identify the problem, define system scope, write functional and non-functional requirements, and study project feasibility.
Weeks 5–8	System Design	Prepare the UI/UX design, system architecture, database design, and main workflow of the platform.
Weeks 8–12	Implementation	Develop the frontend, connect APIs, implement QR logic, and integrate geofencing features.
Weeks 12–14	Testing and Security Audits	Test system functions, check security requirements, validate QR scanning, location checking, and user roles.
Week 15	Documentation	Prepare the final project report, presentation, and project documentation.





Outline

- ☐ Foreword
- ☐ Background
- ☐ Research Problem
- ☐ Contribution
- ☐ Related Work
- ☐ Design and Implementation
- ☐ Results
- ☐ Conclusion
- ☐ Time table
- ☐ **Future Work**

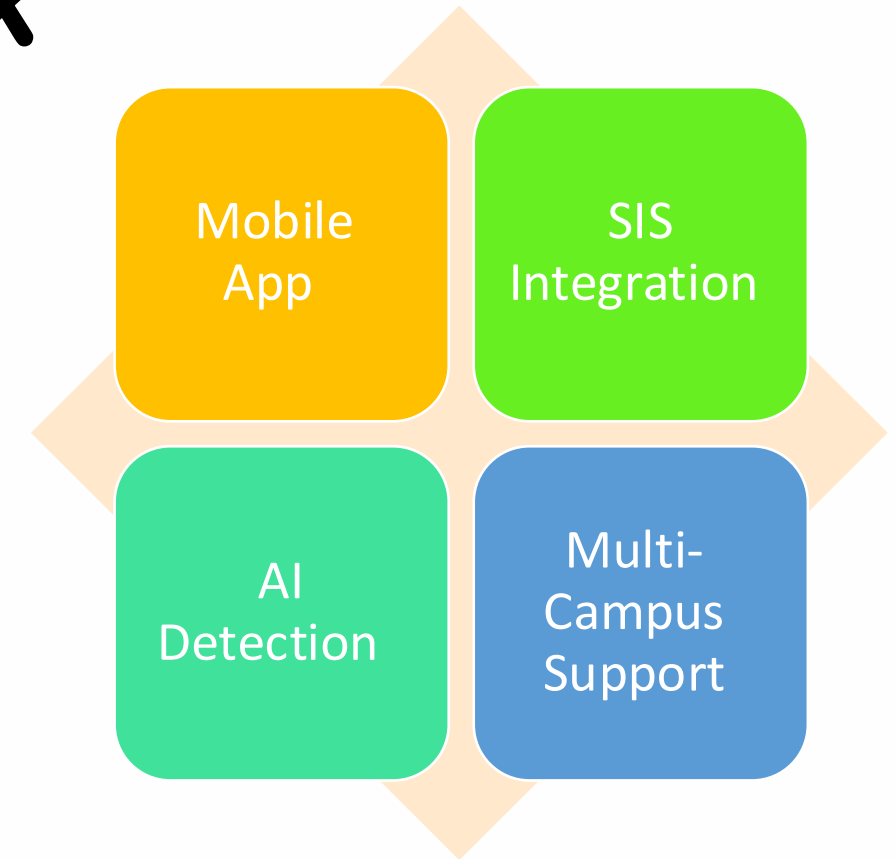




Future Work

Integration & Mobility: Full integration with the university SIS and launching dedicated iOS/Android apps.

- **Advanced AI:** Implementing behavior-based fraud detection and optional face recognition.
- **Technical Enhancements:** Adding offline mode with strong synchronization and bilingual support (Arabic/English).
- **Scalability:** Expanding to multiple campuses and departments with more detailed analytics dashboards.





References

- Alghamdi, S., & Alqarni, A. (2023). Smart attendance systems in higher education: Challenges and opportunities. *International Journal of Advanced Computer Science and Applications*, 14(2), 112–120.
- Alotaibi, M. (2022). Cybersecurity and privacy challenges in educational technology systems. *Journal of Information Security and Cybercrime Research*, 5(1), 45–56.
- Khan, M., & Ibrahim, R. (2021). QR-code-based attendance systems: Security limitations and improvement methods. *International Journal of Computer Applications*, 183(18), 25–31.
- Ministry of Communications and Information Technology. (2023). *Digital transformation in Saudi Arabia and Vision 2030*. Riyadh, Saudi Arabia.
- Saudi Vision 2030. (2024). *Digital transformation and education development goals*. Saudi Vision 2030 Official Portal.
- World Wide Web Consortium. (2018). *Web Content Accessibility Guidelines (WCAG) 2.1*. W3C.





Attendexa

كلية علوم وهندسة الحاسب الآلي
College of Computer Science and Engineering
جامعة حفر الباطن
University of Hafer Al Batin



Questions

جامعة حفر الباطن
University of Hafr Al Batin



